

allowing Swiss-type claim 1 in the set of claims of the main request. Accordingly, the main request was not allowable.

In T 1021/11 too, the main request comprised two independent claims for the same medical indication of the same substance, one claim drafted in the Swiss-type format and the other claim following the provisions in Art. 54(5) EPC. The application had been pending when decision G 2/08 (OJ 2010, 456) was issued and therefore belonged to the category of applications in which the Swiss-type format could, as a general rule, still be used. As both claim formats were available for the application and claims in both formats were present in the main request, the question arose whether both claim types could be present in a single set of claims. The board concurred with the conclusion reached in T 1570/09 that decision G 2/08 did not give applicants an absolute right to draft two independent claims in one single set of claims for one and the same medical indication of one and the same substance, one of those claims being in the Swiss-type format and the other in the format in accordance with Art. 54(5) EPC. However, it appeared to the board that no prohibition of the coexistence of such claims in one claim set could be deduced from G 2/08 either, as it was silent in this respect. Having carefully examined the reasoning given in decision T 1570/09, the board found that several factors prevented it from objecting to the presence of claims in the two formats in one single set of claims in the case before it. Firstly, it noted that a single set of claims could be governed by provisions of the EPC 1973 and the revised EPC at the same time. Secondly, the continued existence of the Swiss-type format, in parallel to the provisions of Art. 54(5) EPC, was a direct consequence of the transitional arrangement provided for by the Enlarged Board in decision G 2/08. Thirdly, the board saw no reason to prevent an applicant from choosing both available formats during the interim period and considered it justified to do so in one set of claims. Even though the claims in both formats provided patent protection for the same medical indication, there was a difference in the subject-matter of the claims due to their category, in combination with their technical features (see above T 1780/12, T 879/12). Thus, by filing two patent applications having the same effective date (two parallel applications or parent/divisional or priority/subsequent application) it was possible for an applicant to obtain patent protection for the same second or further medical indication in both available claim formats. The board therefore did not object to the presence of both formats in a single set of claims, as both formats were applicable to the present application. The board noted that no objections had been raised in similar previous cases (see T 396/09 and T 1869/11), even if the issue had not been discussed in those decisions.

7.2.4 Novelty of the therapeutic application

a) General

The cases decided following decision G 5/83 throw some light on the categories of novel and inventive therapeutic use for which the manufacture of a known substance or compound may be considered patentable. Manufacture of a known composition has been considered patentable for use in a new therapy where the target group to be treated was different (seronegative pigs instead of seropositive pigs; T 19/86, OJ 1989, 25), and in the case of a new therapy with a different technical effect (prevention of tooth decay by means

of a known substance, but by removing plaque instead of by reducing the solubility of tooth enamel; T.290/86, OJ 1992, 414) or a new therapy with a different mode of administration (subcutaneous instead of intramuscular injection; T.51/93).

b) New therapeutic application based on the group of subjects to be treated

According to the established case law of the boards of appeal, the use of the same compound in the treatment of the same disease for a particular group of subjects, can nevertheless constitute a novel therapeutic application, provided that it is carried out on a new group of subjects which is distinguished from the former by its physiological or pathological status (see T.19/86, OJ 1989, 24; T.893/90, T.233/96, T.1399/04, T.734/12).

In T.19/86 (OJ 1989, 25) the board had to decide whether the application of a known medicament for the prophylactic treatment of the **same disease** in an **immunologically different** population of animals of the same species could be considered a new therapeutic application from which novelty for the claims could be derived. According to decision T.19/86 the question of whether a new therapeutic use was in accordance with decision G.5/83 should not be answered exclusively on the basis of the ailment to be cured but also on the basis of the subject (in the case in question, the new group of pigs) to be treated. A therapeutic application was incomplete if the subject to be treated was not identified; only a disclosure of both the disease **and** the subject to be treated represented a complete technical teaching. The proposal according to the application to protect animals which could not hitherto be protected from the disease in question, by intranasally administering to them a known serum, could not be considered disclosed in the prior art and therefore constituted a novel therapeutic application in accordance with G.5/83.

In T.233/96 the board held that if the use of a compound was known in the treatment or diagnosis of a disease of a particular group of subjects, the treatment or diagnosis of the same disease with the same compound could nevertheless represent a novel therapeutic or diagnostic application, provided that it was carried out on a new group of subjects which was **distinguished** from the former **by its physiological or pathological status** (T.19/86, OJ 1989, 25; T.893/90). This did not apply, however if the group chosen overlapped with the group previously treated, or the choice of the novel group was arbitrary which meant that no functional relationship existed between the particular physiological or pathological status of this group of subjects (here humans who were unable to exercise adequately) and the therapeutic or pharmacological effect achieved.

In T.694/16 the board held that if a claim is directed to a known compound or composition for use in a therapeutic method of treatment or prevention of a disease, and the claim specifies that the subject to be treated displays a clearly defined and detectable marker, which is not displayed by all subjects affected by or likely to develop that disease, then the purposive selection of the patients displaying the marker for the specified treatment is a functional feature characterising the claim. When reading the claim in a technically sensible and constructive manner, the skilled person would promptly understand that the purpose of the treatment is to target selectively prodromal patients identified by the CSF markers, rather than other subjects that do not display the markers. This implies that there is a functional relationship between the markers that characterise the patients and the

therapeutic effect which is sought. The presence of this functional relationship confirms that the purposive selection of patients is an essential technical feature qualifying the claim. This has to be taken into account when assessing novelty.

In T 1991/17, a compound for use in the treatment or prevention of bone metabolic diseases associated with osteopenia by inducing osteogenesis was claimed. The respondent had relied on two aspects that would constitute a new specific use of a known compound in a method of treatment: 1) the term "by inducing osteogenesis" as a technical effect and 2) the sub-group of patients to be treated, distinguished on the basis of this technical effect. The term "by inducing osteogenesis" did not form part of the definition of the disease, but qualified the treatment of the disease. The question arose of whether it was to be considered a mere mechanistic explanation of the treatment, or whether this feature linked the treatment to a physiological effect suitable for establishing novelty. The board considered that the patent in suit provided information on a new mechanism of action of peptide D, one of the claimed compounds. This mechanism of action was closely and inseparably linked to the known activity of peptide D of inhibiting bone resorption. No evidence of complete uncoupling of these two activities had been shown. The claimed use could not be distinguished from the known use of peptide D. The technical effect of inducing osteogenesis was therefore not apt to constitute a new specific use.

In T 2232/17 the board did not see any evidence that patients suffering from MS and who are "dependent on steroids" differed in their physiological or pathological status from the general group of MS patients, in particular with respect to their suitability for being treated with natalizumab. The patent contained no data on the treatment of MS patients with steroids and/or natalizumab. Moreover, even if the patent were taken as disclosing a sub-group of "steroid dependent" patients, no special therapeutic or pharmacological effect of natalizumab in this sub-group was disclosed in the patent.

c) New technical effect

In decision T 290/86 (OJ 1992, 414) the board considered the claimed invention new. The grounds for its decision were as follows: "When a prior document and a claimed invention are both concerned with a similar treatment of the human body for the same therapeutic purpose, the claimed invention represents a further medical indication as compared to the prior document within the meaning of decision G 5/83 if it is based upon a different technical effect which is both new and inventive over the disclosure of the prior document". In this case the technical effect considered new was the removal of dental plaque, whereas the prior art only disclosed the depression of enamel solubility in organic acids (see also T 542/96 and T 509/04).

In T 836/01 the board accepted that claims directed to the use of IL-6 to directly influence tumour growth and differentiation were novel over a prior art disclosure of the use of IL-6 to indirectly treat cancer by activating T cells, finding that a new technical effect resided in the medical indication of the treatment of cancer vs. enhancement of the immune system. Applying the principles of decision G 5/83 (OJ 1985, 64), the board concluded that the technical effect relied upon in the claimed invention identified a **new clinical situation**. Since a new clinical situation was inseparable, as an abstract concept, from a patient

suffering under it, it had to be concluded that this new clinical situation also identified a new sub-group of subjects being treated (T.1642/06).

In T.2251/14 the board referred to T.836/01 and stated that a second medical use of a known substance can only be considered novel if the new technical effect leads to a truly new industrial/commercial application arising from the healing of a different clinical situation. It held that in the case in hand, the new technical effect of "enhancing the balance of beneficial and deleterious bacteria in the gastrointestinal tract of an animal having or at risk for inflammatory bowel disease (IBD)" defined a new medical use, since the claimed new technical effect identified a new clinical situation, namely one where it was possible to target the gastrointestinal flora of an animal having or at risk for IBD.

In T.1955/09, the board needed to decide whether the use claimed represented a further and different therapeutic use from the disclosure in document (D1). It stated that the conclusion could not be drawn that the technical effect relied upon by the claimed invention, i.e. the antibiotic effect, was a mere explanation of how the compounds inhibited or neutralised toxins. Rather, this effect identified a new clinical situation, namely one in which it could be preferable to target the infection itself, not merely the toxins produced by the bacteria or fungi causing the infection. The board decided that this reasoning was based on the differentiation of a direct and indirect effect on claims relating to second or further medical uses of a known substance (see above T.836/01 and T.1642/06). In view of the foregoing, the board was satisfied that the subject-matter of claim 1 at issue fulfilled the requirements of Art. 54(1) and (3) EPC vis-à-vis the disclosure in document D1.

In T.1972/14, the claim in question was a Swiss-type claim directed to the preparation of a certain infant formula so as to continuously reduce the circulating level of IGF-1 in the first few months of the life of an infant and thereby reduce the risk of development of obesity later in life. Reducing the risk of developing obesity later in life represented the therapeutic effect to be achieved, while the continuous reduction of the circulating level of IGF-1 represented the mechanism underlying this effect. A Swiss-type claim can derive novelty from the claimed therapeutic effect, but not from the mechanism underlying it. The prior art used the same formulation and the board held that the effect was thereby disclosed since the prior art in question started from the premise that there was a link between reduced protein content and obesity later in life.

d) Same illness

In T.108/09 the claims were drafted in the second medical use format according to decision G 5/83. The board noted that according to decision G 2/08 (OJ 2010, 456), Art. 54(5) EPC does not exclude a medicament already known to be used in the treatment of an illness from being patented for use in a different treatment by therapy of the same illness. It considered it appropriate to evaluate whether the breast cancer of claim 1 as granted was identical to the breast cancer according to document D2. The board noted that the tumours of document D2, being only resistant to tamoxifen, could be distinguished from the tumours of claim 1 as granted, which were additionally resistant to an aromatase inhibitor. This distinction meant that in the case at issue two different diseases or two subsets of a disease (tumour) were concerned. As a consequence, in analogy to the findings in

T. 893/90, the board established novelty. The subject-matter of claim 1 as granted was therefore novel over D2.

e) New dosage regimen

Some boards of appeal have considered recognising specific therapeutic uses as patentable in principle to be problematic where they differ from the prior art merely in terms of the dosage regimen. In G 2/08 (OJ 2010, 456), the Enlarged Board addressed this issue in detail.

With reference to the case law and the danger of a collision with Art. 52(4) EPC 1973, decision T. 584/97 denied patentability for a claim directed essentially to the administration of nicotine in increasing doses. In T. 317/95, T. 56/97 and T. 4/98 (OJ 2002, 139) the issue was discussed, with answers tending towards the negative, but ultimately left undecided. In all of these cases, the grant of a patent would anyway have been refused on other grounds – i.e. lack of novelty or inventive step – so that the outcome of a decision on this issue was immaterial (see also T. 1319/04, OJ 2009, 36)

In T. 1319/04 (OJ 2009, 36), the board referred to the Enlarged Board of Appeal the question whether medicaments for use in methods of treatment by therapy, where the only novel feature was a dosage regime, are patentable under Art. 53(c) and 54(5) EPC. The Enlarged Board answered in G 2/08 (OJ 2010, 456) as follows:

1. Art. 54(5) EPC does not exclude a medicament which is already used in the treatment of an illness being patented for use in a different treatment by therapy of the same illness. Art. 53(c) EPC, which lists as an exception to patentability 'methods for treatment of the human body ... by therapy' is clear and unambiguous and draws a borderline between unallowable method claims directed to a therapeutic treatment on the one hand and allowable claims to products for use in such methods on the other. The two concepts of a method of treatment by therapy and of a product to be used in such a method are so close to each other, that there is a considerable risk of confusion between them unless each is confined to its own domain as allocated to it by the law. Art. 53(c), second sentence, EPC, is not therefore to be interpreted narrowly; on the contrary, it is appropriate to give both provisions (Art. 54(5) and Art. 53(c) EPC) the same weight and conclude that, in respect of claims directed to therapy, method claims are absolutely forbidden in order to leave the physician free to act unfettered, whereas product claims are allowable, provided their subject-matter is new and inventive. By virtue of a legal fiction, Art. 54(4) and (5) EPC acknowledge the notional novelty of substances or compositions even when they are already comprised in the state of the art, provided they are claimed for a new use in a method which Art. 53(c) EPC excludes from patent protection. The notional novelty, and thus, non-obviousness, if any, is not derived from the substance or composition as such, but from its intended therapeutic use. Art. 54(5) EPC refers to "any specific use" [emphasis added] and thus, in conjunction with the stated intention of the legislator to maintain the status quo of the protection evolved in the case law of the boards of appeal under G 5/83 (OJ 1985, 64) in this respect, this use cannot be ex officio limited to a new indication stricto sensu (approving T. 1020/03, OJ 2007, 204).